Group: Date:

Worksheet No. 1

EXPLORING QUADRATIC FUNCTIONS OF THE FORM

$$y = ax^2 + bx + c$$

Read carefully and fill in the blanks:

Task 1: Visualise the curve's shape

Step 1: Scan QR Code below to access GeoGebra Classroom.

Step 2: On the left panel, click the "Algebra" tab (the calculator icon).

Step 3: Click the input bar and type: then press "Enter".

Step 4: Move the sliders for and to any value.

Step 5: Now, focus on slider a. Slide it and observe the change:

- When ....., the graph is $\cup$ -shaped.
- When ....., the graph is $\cap$ -shaped.

Great job!

Group: Date:

Worksheet No. 1

EXPLORING QUADRATIC FUNCTIONS OF THE FORM

$$y = ax^2 + bx + c$$

Read carefully and fill in the blanks:

Task 2: : Consider the graph $y = 2x^2 + 4x - 6$ follow the following steps to find the essential information for sketching the graph:

1. Shape: a =....., so the graph is ... -shaped.
2. y-intercept: Set $x = 0$ then $y =$ ..... Coordinate: (0, ...).
3. x-intercepts: Set $y = 0$ then we obtain $2x^2 + 4x - 6 = 0$

- Factorize: .....$= 0$.
- $x =$ ..... or $x =$ ..... Coordinates: (....., 0) and (....., 0).

4. Turning Point:

- Find axis of symmetry: $x =$.....
- Substitute $x =$..... to find $y =$.....
- Turning point coordinate: (....., ...).

Great job!

Group: Date:

Worksheet No. 1

EXPLORING QUADRATIC FUNCTIONS OF THE FORM

$$y = ax^2 + bx + c$$

Read carefully and fill in the blanks:

Task 3: Receive AI feedback

1. Scan QR Code to open the AI Chatbot.

2. Take a clear photo of BOTH **Task 1** and **Task 2** and upload them to the chatbot.
3. To ask the AI to check your work, please type the prompt: *"Please check my observations in Task 1 and my calculations in Task 2 for* $y = 2x^2 + 4x - 6$*. Give me hints if there are any errors."*
4. Write down the AI's feedback here:

   .....

   .....

   .....

   .....

   .....

Great job!

Group: Date:

Worksheet No. 1

EXPLORING QUADRATIC FUNCTIONS OF THE FORM

$$y = ax^2 + bx + c$$

Read carefully and fill in the blanks:

Task 4: Find the turning point by completing the square

You can find the coordinates of the turning point of a parabola algebraically by changing the equation from $y = ax^2 + bx + c$ *into the form* $y = a(x+p)^2 + q$. *In this form, the turning point is(−p,q).*

For example, let us consider the equation: $y = 2x^2 + 8x - 8$.

Method 1: By formula

Step 1: Identify the coefficients from the standard form $y = ax^2 + bx + c$:

$a =$ ....., $b =$ ....., $c =$.....

Step 2: Find the x-coordinate using the formula $x = -\frac{b}{2a}$: $x = -\frac{.....}{.....} = ...$

Step 3: Find the y-coordinate by substituting the value of x into the original equation:

$y = ... = ...$

Step 4: Based on calculation, the turning point is: (....., ...).

Method 2: By completing the square

Consider the equation $y = 2x^2 + 8x - 8$

This can be rewritten as $y = 2(x + ...)^2 - .....$ by completing the square.

Squaring any value results in an answer that is either positive or 0. This means that for any value of x, the smallest value of $(x + ...)^2$ is .....

This means that the minimum value of $y = 2(x + ...)^2 - .....$ is ..... and that this occurs when $x = ...$

The turning point of the graph $y = 2(x + ...)^2 - .....$ has the coordinates:

(....., ...).

Great job!